Type MS · Describe how an impulse crosses a synapse

Answer 1 of 10 · Question 3 part a.iii · 6 marks

2024 NOV · Paper 02 · extract 1 of 2 · find part a.iii in this mark scheme extract

3 (a)	(i)	Synapse/synaptic cleft;		1
	(ii)	arrow pointing from top to bottom;		1
	(iii)	any six from • vesicles; • contain neurotransmitter/ named neurotransmitter; • (vesicles)fuse/binds/attaches with pre-synaptic membrane/(vesicles) moves to pre-synaptic membrane; • exocytosis/release of neurotransmitter; • which diffuses (across the gap/synapse)/description of diffusion; • binds/attaches with receptors on post-synaptic membrane; • (initiates)new impulse;		6
	(iv)	• neurotransmitter only released from pre-synaptic membrane/ neurone A/first neurone/(vesicles) with neurotransmitter in pre-synaptic knob/neurone A/first neurone; • diffuses down a concentration gradient/description of concentration gradient; • receptors only on post-synaptic membrane/neurone B/ second neurone;		3
(b)		any two from • addiction; • stroke/heart problems/high blood pressure; • steal to fund habit; • anxiety; • depression; • hallucinations; • memory loss; • brain damage/ damage nerve cells; • paranoia; • overdose/death;		2

Total 13 marks

Type MS · Describe how an impulse crosses a synapse

Answer 1 of 10 · Question 3 part a.iii · 6 marks

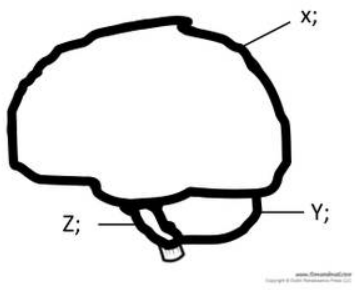
2024 NOV · Paper 02 · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe how an impulse crosses a synapse

Answer 2 of 10 · Question 7 part b.iii · 3 marks

2019 JUN · Paper 01 · extract 1 of 2 · find part b.iii in this mark scheme extract

7 (a)		Letters can be anywhere within outline shown. Allow labels in place of than letters.	3
(b) (i)	Motor;		1
(ii)	Any one from the following: (sensory neurone): longer dendrites/shorter axon/cell body outside of spinal cord/CNS/in middle of neurone/transmit impulse to the CNS/brain/spinal cord (motor neurone) shorter dendrites/longer axon/cell body at one end of neurone/inside spinal cord/transmit impulses from CNS/brain/spinal cord	Answer must be comparative and state which type of neurone is being referred to	1
(iii)	• chemicals/neurotransmitters; • diffuse; • across a synapse; • bind to receptors (on post-synaptic membrane);		3
(c)	• stem cells are undifferentiated/unspecialised/can differentiate; • could be made to form nerve cells; • to replace damaged cells in brain;		1 1 1

Total question 7 = 11 marks

Type MS · Describe how an impulse crosses a synapse

Answer 2 of 10 · Question 7 part b.iii · 3 marks

2019 JUN · Paper 01 · extract 2 of 2 · find part b.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe how an impulse crosses a synapse

Answer 3 of 10 · Question 5 part c.ii · 3 marks

2020 NOV · Paper 01 · extract 1 of 2 · find part c.ii in this mark scheme extract

Number			
5 (a)	any three from • rapid response (to a stimulus) (1) • involuntary/automatic (1) • no brain involvement (1) • protective (1)		3 max
(b)	• A = motor (1) • B = relay/association/connector (1) • C = sensory (1)		3
(c) (i)	• fewer branches/dendrons/dendrites (1) • shorter branches/ dendrons/dendrites (1) • fewer knobs/synapses (1)	R axons	3
(ii)	• transmits impulse (1) • across synapse/(synaptic) cleft (1) • stimulates impulse in next neurone (1)		3
(iii)	• secreted by synapses/synaptic knob (1) • fewer synapses/knobs (1) • less secreted by remaining synapses/knobs (1)		3

Total 15 marks

Type MS · Describe how an impulse crosses a synapse

Answer 3 of 10 · Question 5 part c.ii · 3 marks

2020 NOV · Paper 01 · extract 2 of 2 · find part c.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe how an impulse crosses a synapse

Answer 4 of 10 · Question 8 part c · 6 marks

2021 JUN · Paper 01 · find part c in this mark scheme extract

number				
8	(a)	any three from • rapid response; • automatic; • to protect body; • from harmful effects/named example;		3
	(b)	correct position;		1
	(c)	any six from • impulse arrives at end of axon; • synapse; • causes release of neurotransmitter; • e.g. acetylcholine; • from vesicles; • diffuses across gap/cleft; • combines with/stimulates (post synaptic) receptors; • triggers impulse;		6

Total 10 marks

Type MS · Describe how an impulse crosses a synapse

Answer 5 of 10 · Question 6 part b.ii · 4 marks

2022 JAN · Paper 02 · extract 1 of 2 · find part b.ii in this mark scheme extract

Number				
6	(a)	<table><tr><th>Nervous system</th><th>Hormonal system</th></tr><tr><td>nerve fibres/cells;</td><td>blood;</td></tr><tr><td>electrical/electrochemical;</td><td>chemical;</td></tr><tr><td>fast;</td><td>slow;</td></tr></table>	Allow nerves / neurones Reject hormone	3
	(b) (i)	arrow drawn left to right;		1
	(ii)	• neurotransmitter; • released from synapse/synaptic knobs; • diffuses across gap; • bind to receptors on next cell;		4
	(iii)	• neurotransmitter only released at one side of synapse; • can only diffuse in one direction; • from high to low concentration; • receptors only present on post synaptic membrane;		4

Total 12 marks

Type MS · Describe how an impulse crosses a synapse

Answer 5 of 10 · Question 6 part b.ii · 4 marks

2022 JAN · Paper 02 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe how an impulse crosses a synapse

Answer 6 of 10 · Question 3 part b.i · 1 mark

2022 JUN · Paper 02 · find part b.i in this mark scheme extract

3 (a)	In the following order: • receptors; • sensory; • relay; • motor;		4
(b) (i)	synapse;		1
(ii)	• chemicals/neurotransmitters; • diffuse across synaptic clef/gap;		2
(c)	• 60 × 5; • 300 x 120 = 36 000;		2
			Total 9

Question number	Answer	Notes	Marks

Type MS · Describe how an impulse crosses a synapse

Answer 7 of 10 · Question 3 part b.ii · 2 marks

2022 JUN · Paper 02 · find part b.ii in this mark scheme extract

3 (a)	In the following order: • receptors; • sensory; • relay; • motor;		4
(b) (i)	synapse;		1
(ii)	• chemicals/neurotransmitters; • diffuse across synaptic clef/gap;		2
(c)	• 60 × 5; • 300 x 120 = 36 000;		2
			Total 9

Question number	Answer	Notes	Marks

Type MS · Describe how an impulse crosses a synapse

Answer 8 of 10 · Question 5 part a.i · 2 marks

2023 JUN · Paper 02 (regional R) · find part a.i in this mark scheme extract

Number				
5	(a) (i)	Any two from: • chemical; • diffused from one neurone to another; • allows neurones to communicate / reference to communication between presynaptic and post synaptic neurones;		2
	(ii)	• neurotransmitter/dopamine/ diffuses/travels across synapse; • binds with receptor (on postsynaptic membrane)		2
	(iii)	Any two from: • increased movement; • increased heart rate/blood pressure; • more energy; • sleeping difficulties; • anxiety/stress; • hallucinations;		2
	(b) (i)	Any two from • tremors/random muscular movement / involuntary movement; • slower movement; • speech/writing changes; • impaired posture/balance; • sleeping problems; • rigidity/stiffness; • anxiety/depression/mental health/memory problems;		2
	(ii)	• (145 000 ÷ 100) (x 18) = 26 100; • (26 100 + 145 000) = 171 100;	Allow ecf for mp2 - if mp1 has an incorrect value but this value is added to 145 000 to arrive at the correct answer for the calculation award 1 mark Full marks for correct final answer	2

Total for question 5 = 10 marks

Type MS · Describe how an impulse crosses a synapse

Answer 9 of 10 · Question 5 part a.ii · 2 marks

2023 JUN · Paper 02 (regional R) · find part a.ii in this mark scheme extract

Number				
5	(a) (i)	Any two from: • chemical; • diffused from one neurone to another; • allows neurones to communicate / reference to communication between presynaptic and post synaptic neurones;		2
	(ii)	• neurotransmitter/dopamine/ diffuses/travels across synapse; • binds with receptor (on postsynaptic membrane)		2
	(iii)	Any two from: • increased movement; • increased heart rate/blood pressure; • more energy; • sleeping difficulties; • anxiety/stress; • hallucinations;		2
	(b) (i)	Any two from • tremors/random muscular movement / involuntary movement; • slower movement; • speech/writing changes; • impaired posture/balance; • sleeping problems; • rigidity/stiffness; • anxiety/depression/mental health/memory problems;		2
	(ii)	• (145 000 ÷ 100) (x 18) = 26 100; • (26 100 + 145 000) = 171 100;	Allow ecf for mp2 - if mp1 has an incorrect value but this value is added to 145 000 to arrive at the correct answer for the calculation award 1 mark Full marks for correct final answer	2

Total for question 5 = 10 marks

Type MS · Describe how an impulse crosses a synapse

Answer 10 of 10 · Question 3 part a.i · 1 mark

2024 NOV · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

3	(a)	(i)	Synapse/synaptic cleft;		1
		(ii)	arrow pointing from top to bottom;		1
		(iii)	any six from • vesicles; • contain neurotransmitter/ named neurotransmitter; • (vesicles)fuse/binds/attaches with pre-synaptic membrane/(vesicles) moves to pre-synaptic membrane; • exocytosis/release of neurotransmitter; • which diffuses (across the gap/synapse)/description of diffusion; • binds/attaches with receptors on post-synaptic membrane; • (initiates)new impulse;		6
		(iv)	• neurotransmitter only released from pre-synaptic membrane/ neurone A/first neurone/(vesicles) with neurotransmitter in pre-synaptic knob/neurone A/first neurone; • diffuses down a concentration gradient/description of concentration gradient; • receptors only on post-synaptic membrane/neurone B/ second neurone;		3
	(b)		any two from • addiction; • stroke/heart problems/high blood pressure; • steal to fund habit; • anxiety; • depression; • hallucinations; • memory loss; • brain damage/ damage nerve cells; • paranoia; • overdose/death;		2

Total 13 marks

Type MS · Describe how an impulse crosses a synapse

Answer 10 of 10 · Question 3 part a.i · 1 mark

2024 NOV · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks